Lecture 6: Atmospheres II
Clouds, Weather, & Climate

Introduction to Planetary Science

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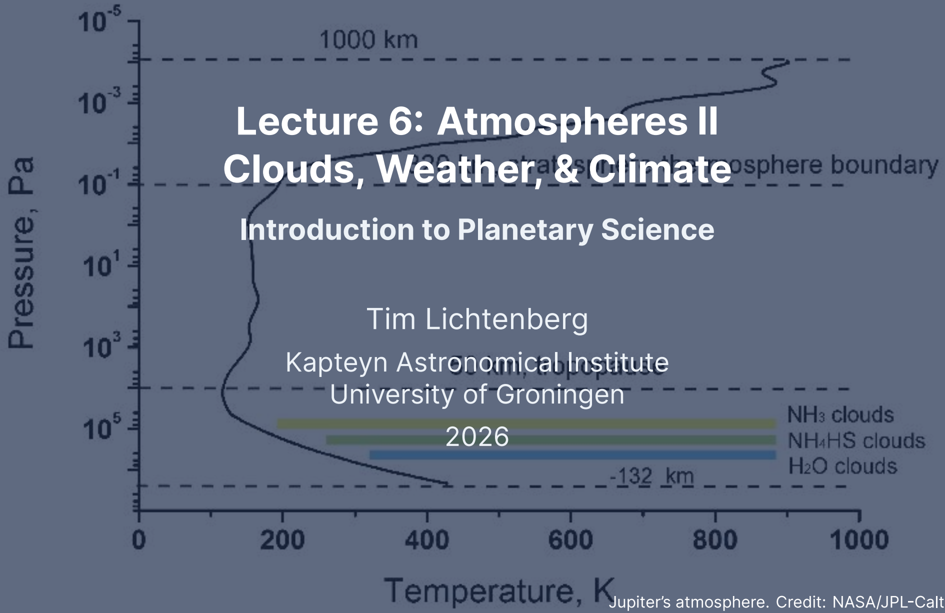


Jupiter's atmosphere. Credit: NASA/JPL-Caltech/SwRI/MSSS

Learning Objectives

By the end of this lecture, you will be able to:

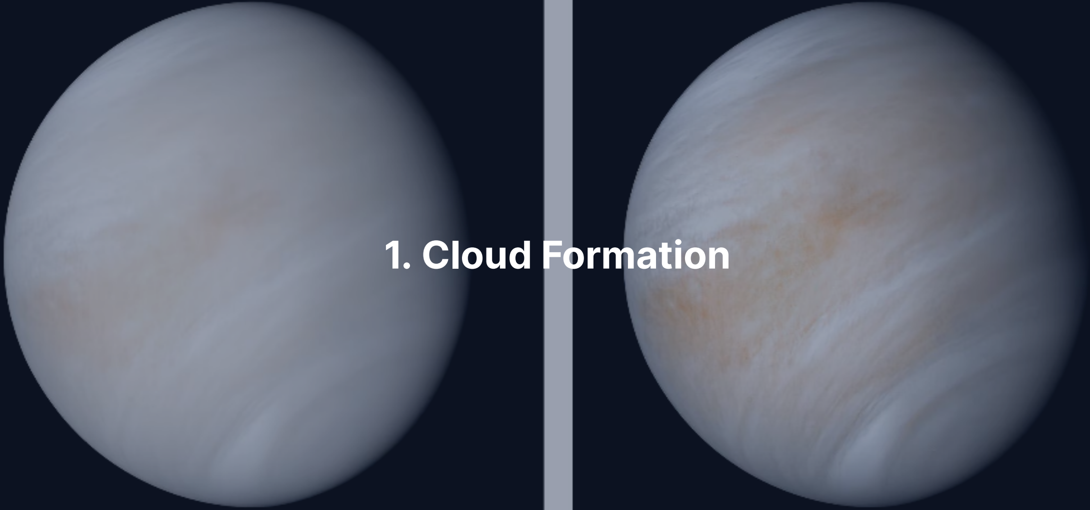
- Derive the Clausius-Clapeyron equation and apply it to predict cloud formation
- Explain atmospheric circulation using the Coriolis effect and geostrophic balance
- Describe weather phenomena across the solar system
- Discuss long-term climate evolution, including the faint young Sun paradox
- Explain the carbonate-silicate cycle as a climate thermostat

Recap: Lecture 5

- Atmospheric types: primary ($\mathrm{H_2/He}$), secondary (outgassed), tertiary (modified)
- Hydrostatic equilibrium: $\mathrm{d}P = -\rho g\,\mathrm{d}z$
- Scale height: $H = k_\mathrm{B}T/(\mu\, m_u\, g)$
- Radiative equilibrium: $T_\mathrm{eq} = T_\odot(R_\odot/2a)^{1/2}(1 - A)^{1/4}$
- Greenhouse effect: surface temperature exceeds equilibrium temperature
- Adiabatic lapse rate: $\Gamma_d = g/c_p$

Today: What happens when vapour condenses? How do winds blow? Why has Earth's climate been stable for 4 Gyr?

1. Cloud Formation



Venus in UV, Mariner 10 (1974). Credit: NASA/JPL-Caltech

Saturation and Condensation

- Every vapour has a **saturation vapour pressure** $P_{\mathrm{sat}}(T)$
- When actual vapour pressure exceeds P_{sat}: air is **supersaturated**
- $P_{\mathrm{sat}}(T)$ increases roughly exponentially with temperature
- **Relative humidity:**

$$\mathrm{RH} = \frac{P_{\mathrm{vapour}}}{P_{\mathrm{sat}}(T)} \times 100\%$$

- $\mathrm{RH} = 100\%$: saturated; $\mathrm{RH} > 100\%$: condensation can occur
- Rising air cools → P_{sat} drops → reaches saturation → clouds

Nucleation: How Droplets Form

Even in supersaturated air, condensation needs a **seed**.

Homogeneous nucleation: Forming droplets from vapour alone
Requires very high supersaturations ($\mathrm{RH} \gg 100\%$), extremely rare

Heterogeneous nucleation: Condensation onto pre-existing particles
Condensation nuclei: dust, volcanic aerosols, sea salt, soot
Occurs at modest supersaturation ($\mathrm{RH} \gtrsim 100\%$): the dominant mechanism

- Earth: oceanic and biogenic nuclei
- Mars: wind-lofted mineral dust
- Giant planets: photochemical hazes from above

Lifting Condensation Level (LCL)

- Rising air cools at the dry adiabatic lapse rate $\Gamma_d = g / c_p$
- Vapour pressure stays roughly constant during ascent
- But $P_{\mathrm{sat}}(T)$ decreases as temperature drops
- At the altitude where $P_{\mathrm{vapour}} = P_{\mathrm{sat}}(T)$: condensation begins

The **lifting condensation level** (LCL) marks the **cloud base**. Above it, latent heat release produces the moist adiabatic lapse rate (~5–6 K km^{-1} on Earth vs. ~9.8 K km^{-1} dry).

Latent heat → buoyancy → vigorous convection → thunderstorms, hurricanes, cumulonimbus.

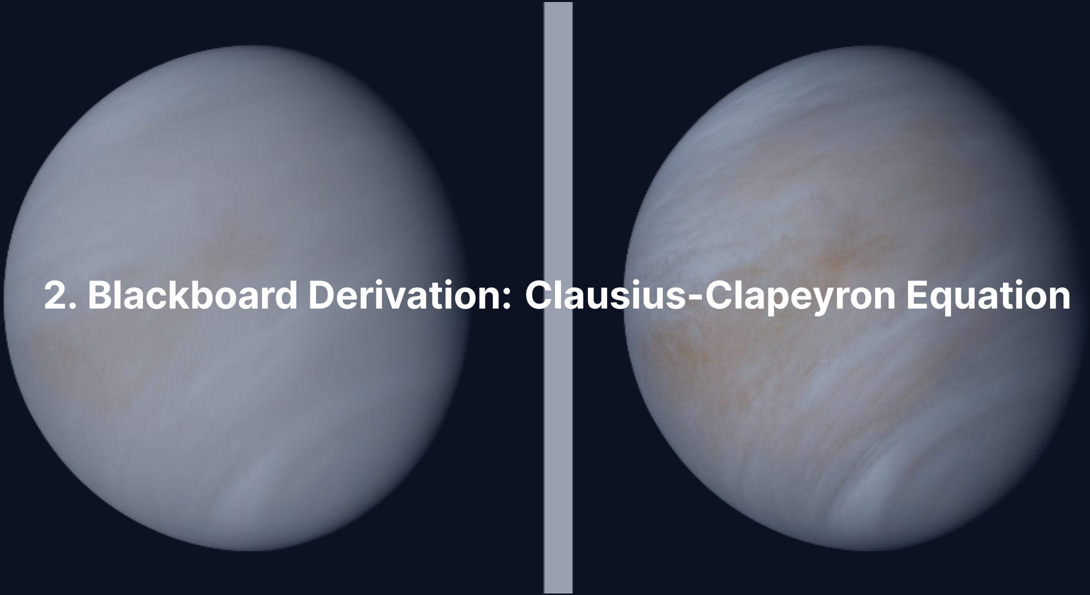
Cloud Types Across the Solar System

What condenses depends on temperature and composition:

Body	Cloud species	Cloud base altitude
Earth	H_2O (liquid & ice)	~1–2 km
Venus	H_2SO_4 (sulfuric acid)	~48 km
Mars	CO_2 ice, H_2O ice	10–100 km
Titan	CH_4, C_2H_6	~8–30 km
Jupiter	NH_3, NH_4SH, H_2O	layered, 0.5–7 bar

The physics is the same everywhere: the **Clausius-Clapeyron equation** governs all of them.

2. Blackboard Derivation: Clausius-Clapeyron Equation



Venus in UV. Credit: NASA/JPL-Caltech

Deriving the Clausius-Clapeyron Equation

Goal: Derive the exponential dependence of saturation vapour pressure on temperature from thermodynamic phase equilibrium.

Setup: phase equilibrium.

Along the coexistence curve, liquid and vapour are in equilibrium:

$$g_\ell(T, P) = g_v(T, P)$$

where g is the specific Gibbs free energy.

Moving along the coexistence curve by $(\mathrm{d}T, \mathrm{d}P)$:

$$\mathrm{d}g_\ell = \mathrm{d}g_v$$

Derivation: Exact Clausius-Clapeyron

From thermodynamics: $\mathrm{d}g = -s\,\mathrm{d}T + v\,\mathrm{d}P$

Setting $\mathrm{d}g_\ell = \mathrm{d}g_v$:

$$(s_v - s_\ell)\,\mathrm{d}T = (v_v - v_\ell)\,\mathrm{d}P$$

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\Delta s}{\Delta v} = \frac{L_v}{T(v_v - v_\ell)}$$

Two approximations:

1. $v_v \gg v_\ell$, so $v_v - v_\ell \approx v_v$
2. Ideal gas: $v_v = R_v T / P$, where $R_v = R^* / M$

$$\frac{\mathrm{d}P_{\mathrm{sat}}}{\mathrm{d}T} = \frac{L_v P_{\mathrm{sat}}}{R_v T^2}$$

Integrating the Differential Form

Starting from

$$\frac{\mathrm{d}P_{\mathrm{sat}}}{\mathrm{d}T} = \frac{L_v\,P_{\mathrm{sat}}}{R_v\,T^2}$$

separate variables:

$$\frac{\mathrm{d}P_{\mathrm{sat}}}{P_{\mathrm{sat}}} = \frac{L_v}{R_v}\,\frac{\mathrm{d}T}{T^2}$$

Integrate from a reference state $(T_{\mathrm{ref}}, P_{\mathrm{ref}})$ to (T, P_{sat}), assuming $L_v \approx \mathrm{const.}$:

$$\ln\left(\frac{P_{\mathrm{sat}}}{P_{\mathrm{ref}}}\right) = -\frac{L_v}{R_v}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)$$

Exponentiate to get the integrated form on the next slide.

The Clausius-Clapeyron Equation

$$\boxed{P_{\mathrm{sat}}(T) = P_{\mathrm{ref}} \exp\left[-\frac{L_v}{R_v}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right]}$$

Credit: Derivation follows Catling (2017), *Atmospheric Evolution on Inhabited and Lifeless Worlds*, Ch. 1

- P_{sat} depends **exponentially** on temperature
- The sensitivity is controlled by L_v/R_v (units of K)
- Large L_v/R_v → very sensitive to temperature changes
- This equation applies to **any** vapour ↔ condensed phase transition

Worked Example: Water on Earth

For water vapour:

L_v	2.50×10^6 J kg^{-1}
M	0.018 kg mol^{-1}
$R_v = R^*/M$	462 J kg^{-1} K^{-1}
T_{ref} (triple point)	273 K
P_{ref} (triple point)	611 Pa
L_v/R_v	$\approx$ 5400 K

At T = 293 K (20°C):

$$P_{\mathrm{sat}} = 611 \exp\left[-5400\left(\frac{1}{293} - \frac{1}{273}\right)\right] = 611 \times 3.86 \approx 2360 \text{ Pa}$$

Measured value: 2300 Pa; excellent agreement.

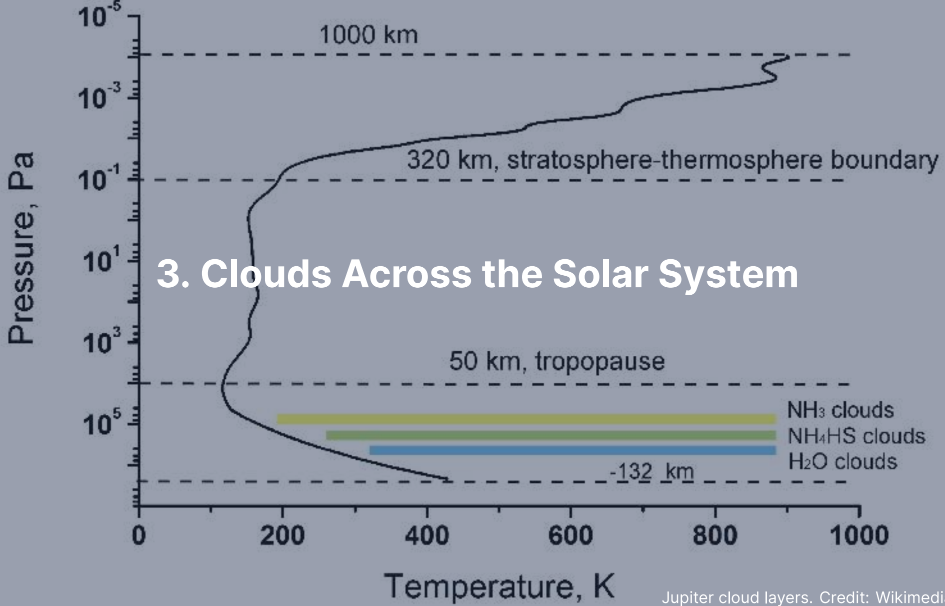
Condensing Species Across the Solar System

Species	L_v (kJ kg^{-1})	R_v (J kg^{-1} K^{-1})	L_v/R_v (K)	T_{cond} (K)	Where
H_2O	2500	462	5400	200–280	Earth, Mars
H_2SO_4	540	85	6400	340–380	Venus
NH_3	1370	488	2800	130–150	Jupiter, Saturn
CH_4	510	519	980	80–90	Titan
CO_2	571	189	3020	145–175	Mars

Source: Catling (2017), de Pater & Lissauer (2010); T_{cond} is typical cloud-base range

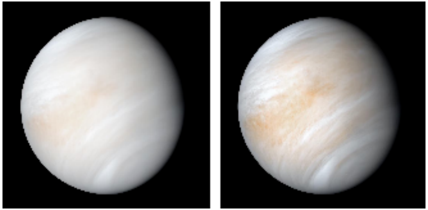
- High L_v/R_v for H_2SO_4: Venus's clouds confined to a narrow altitude range
- Low L_v/R_v for CH_4: Titan's methane clouds extend over a wide range

3. Clouds Across the Solar System



Jupiter cloud layers. Credit: Wikimedia,

Venus: Sulfuric Acid Clouds



Credit: NASA/JPL-Caltech (1974)

- H_2SO_4 clouds from 48 to 70 km altitude
- Completely obscure the surface at visible wavelengths
- Photochemistry:

$$SO_2 + H_2O \xrightarrow{h\nu} H_2SO_4$$

- SO_2 supplied by volcanic outgassing
- Unidentified **UV absorber** creates banded patterns
- Sub-cloud haze extends to ~30 km

Mars Recurring Slope Lineae (RSL)

Credit: NASA/JPL-Caltech/UA/MRO HiRISE

First seen by HiRISE in 2011: narrow dark streaks that grow during the warmest months.

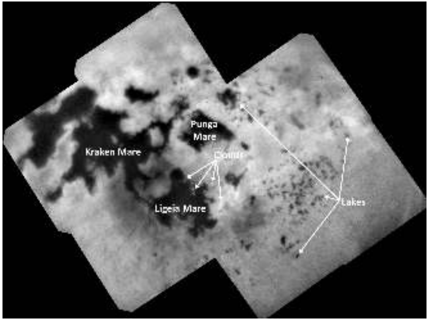
- Found on steep, equator-facing slopes
- Appear in spring/summer, fade in winter
- Length: 50–300 m; width: 0.5–5 m
- **Initial interpretation:** briny water flow (perchlorate salts depress freezing point)
- **Revised view (2017+):** likely **dry granular flow**; spectroscopic water signal re-interpreted

Mars: Dust and Ice Clouds

Mars's thin atmosphere (~6 mbar) supports two cloud types:

- CO_2 **ice clouds:** High altitude (50–100 km), frost point ~148 K
- H_2O **ice clouds:** Lower altitude (10–30 km), over Tharsis region, aphelion cloud belt
- **Mineral dust** plays a central role:
  - Condensation nuclei for ice clouds
  - Absorbs solar radiation, heats atmosphere
  - Positive feedback: more dust → more heating → stronger winds → more dust

Titan's Methane Cycle: An Earthlike Analogue



Credit: NASA/JPL-Caltech/ASI/USGS (Cassini)

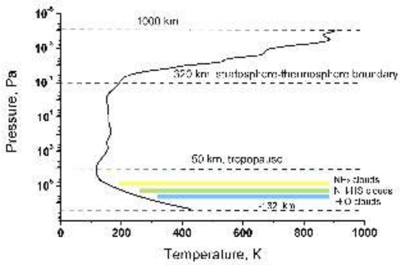
Titan has the only other active hydrological cycle in the solar system, with methane replacing water.

- **Surface:** liquid methane seas near the north pole
- **Kraken Mare:** ~1 100,000 km^2, ~300 m deep
- Rivers carve channels; methane rain fills lakebeds
- Evaporation at equator, rain at poles
- 30-yr Titan year drives slow seasonal cycling
- Cassini radar mapped lakes; Dragonfly will sample in 2034+

Titan: Methane Rain

- Only known active **hydrological cycle** beyond Earth, with $\mathrm{CH_4}$ replacing $\mathrm{H_2O}$
- Surface: $T \approx 94$ K, $P \approx 1.5$ bar, near the triple point of methane
- Liquid methane: lakes, seas, rivers carved into the icy surface
- $\mathrm{CH_4}$ clouds at 8–30 km; $\mathrm{C_2H_6}$ (ethane) clouds also present
- Rainfall infrequent but intense
- Seasonal cloud migration observed by Cassini–Huygens

Giant Planets: Layered Cloud Structure



Credit: Wikimedia, CC BY-SA 3.0

Vertically layered, predicted by Clausius-Clapeyron:

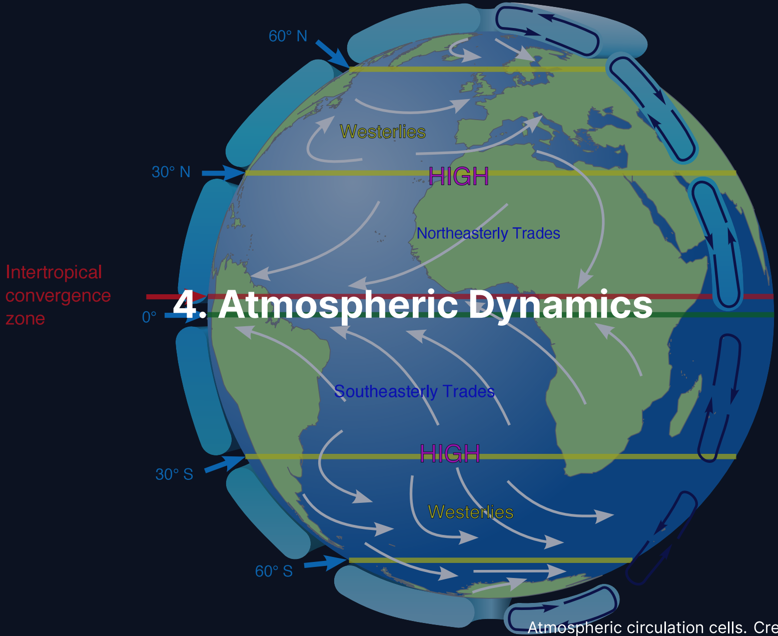
1. $\mathrm{NH_3}$ **ice** (top): T ~ 130–150 K, P ~ 0.5–1 bar
   The visible white and coloured bands
2. $\mathrm{NH_4SH}$: T ~ 200–240 K, P ~ 2–3 bar
   $\mathrm{NH_3} + \mathrm{H_2S} \rightarrow \mathrm{NH_4SH}$
3. $\mathrm{H_2O}$ **ice/liquid** (deep): T ~ 270–300 K, P ~ 5–7 bar
   Drives weather via latent heat release

Ice giants: $\mathrm{CH_4}$ on top, $\mathrm{H_2S}$ below, $\mathrm{NH_4SH}$ and $\mathrm{H_2O}$ deeper.

Break

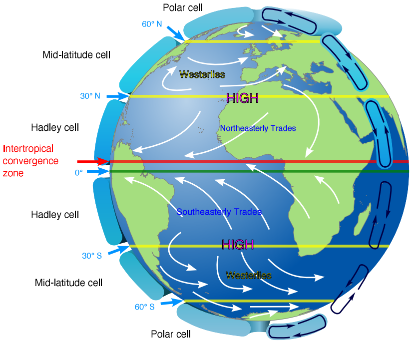
Earthrise, Apollo 8 (1968). Credit: NASA

4. Atmospheric Dynamics



Atmospheric circulation cells. Credit: Wikimedia, public domain

The Hadley Cell



Credit: Wikimedia, public domain

1. Intense solar heating at equator → air rises
2. At altitude, air flows poleward
3. Air cools radiatively, sinks at ~30° latitude
4. Surface return flow → **trade winds**

This loop is the **Hadley cell**:

- Dominant circulation in the tropics
- Primary heat transport: equator → poles
- Sinking zones → desert belts on Earth

The Coriolis Effect

On a rotating planet, moving air is **deflected**:

- **Right** in the Northern Hemisphere
- **Left** in the Southern Hemisphere

The **Coriolis parameter**:

$$f = 2\Omega \sin \phi$$

- Earth: $\Omega = 7.27 \times 10^{-5}$ rad s^{-1}
- At mid-latitudes ($\phi = 45°$): $f \approx 1.03 \times 10^{-4}$ s^{-1}
- At equator ($\phi = 0°$): $f = 0$, so no Coriolis deflection

> The Coriolis effect explains why winds do not blow directly from high to low pressure, and why weather systems rotate.

The Rossby Number

Does rotation matter for a given flow?

$$\mathrm{Ro} = \frac{U}{fL}$$

$\mathrm{Ro} \ll 1$: Rotation dominates: large-scale circulation, jet streams
Earth: $U \sim 10\ \mathrm{m\,s^{-1}}$, $L \sim 1000\ \mathrm{km} \rightarrow \mathrm{Ro} \sim 0.1$

$\mathrm{Ro} \gg 1$: Rotation unimportant: tornadoes, dust devils

Determines the number of circulation cells on a planet.

Circulation Cells and Rotation Rate

Slowly rotating (Venus, Titan):

Single Hadley cell from equator to pole in each hemisphere.
Venus: 243-day period → $\mathrm{Ro} \gg 1$ for large-scale flows.

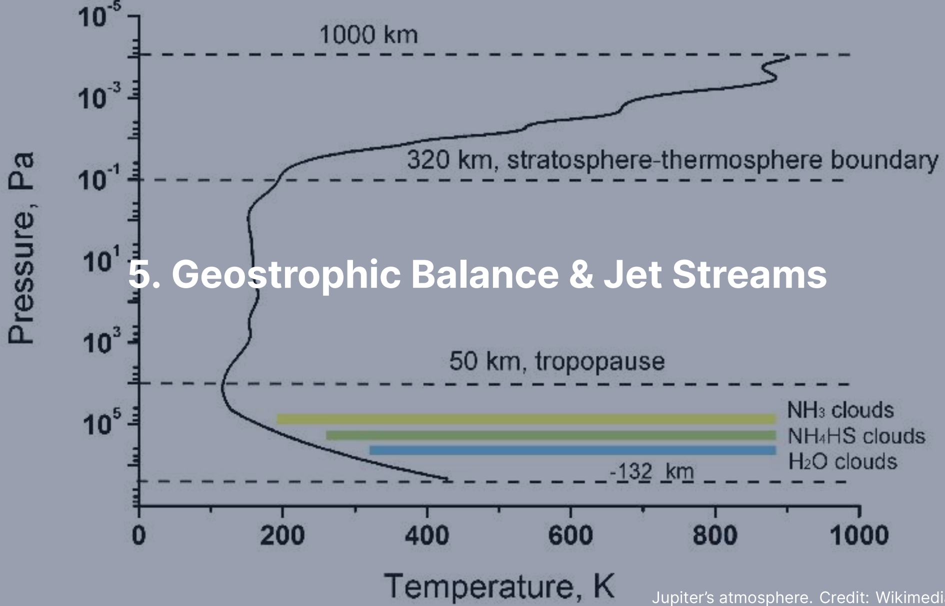
Moderately rotating (Earth):

Hadley cell to ~30°, then Ferrel cell (mid-latitudes), polar cell.
Three cells per hemisphere.

Rapidly rotating (Jupiter, Saturn):

Many alternating cells → **banded structure**.
Jupiter: ~15 alternating jet streams per hemisphere.
Light zones (rising air, high clouds) and dark belts (sinking air).

5. Geostrophic Balance & Jet Streams



Jupiter’s atmosphere. Credit: Wikimedia,

Geostrophic Balance

At large scales ($\mathrm{Ro} \ll 1$), steady state = Coriolis force balances pressure gradient:

$$f\,\hat{k} \times \mathbf{v}_g = -\frac{1}{\rho}\nabla P$$

In scalar form:

$$u_g = -\frac{1}{f\rho}\frac{\partial P}{\partial y}, \qquad v_g = \frac{1}{f\rho}\frac{\partial P}{\partial x}$$

The geostrophic wind blows **parallel to isobars**, not from high to low pressure. In the Northern Hemisphere: low pressure to the left of the wind.

This is why cyclones and anticyclones **rotate** around pressure centres.

Jet Streams and the Thermal Wind

Jet streams: Narrow bands of fast winds (~30–70 m s^{-1} on Earth).

The **thermal wind relation** connects vertical wind shear to horizontal temperature gradients:

$$\frac{\partial \mathbf{v}_g}{\partial z} \propto \hat{k} \times \nabla T$$

- Where ∇T is steepest (tropics vs. poles boundary): wind speed increases with altitude
- On Earth: subtropical jet (~30°) and polar jet (~60°)
- Jet streams **steer weather systems** across the planet

Giant Planet Banding

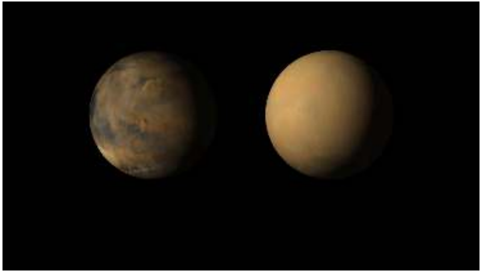
- **Zones** (light): rising air, high clouds (ammonia ice tops)
- **Belts** (dark): sinking air, deeper cloud layers exposed
- Strong **zonal jets** between adjacent bands:
  - Jupiter: ~150 $m\,s^{-1}$
  - Saturn: ~400 $m\,s^{-1}$
- Jets remarkably stable over decades of observation
- Juno gravity measurements: jets extend deep into the interior

The same geostrophic and thermal wind physics that produces Earth's jet streams creates the banded structure of the giant planets, just with many more cells due to rapid rotation.

6. Weather & Storms Across the Solar System

Jupiter’s Great Red Spot. Credit: NASA/JPL-Caltech/SwRI/MSSS

Mars: Dust Storms



Credit: NASA/JPL-Caltech/MSSS

- Local storms: dust lofted to 10–30 km
- Occasionally grow into **global dust storms**
- 2018 global storm ended *Opportunity*'s 15-year mission
- **Positive feedback loop:** dust absorbs sunlight → heats air → stronger winds → more dust

Seasonal $\mathrm{CO_2}$ cycle:

- ~25–30% of atmosphere freezes onto polar caps each winter
- Surface pressure swings: 5–7 mbar

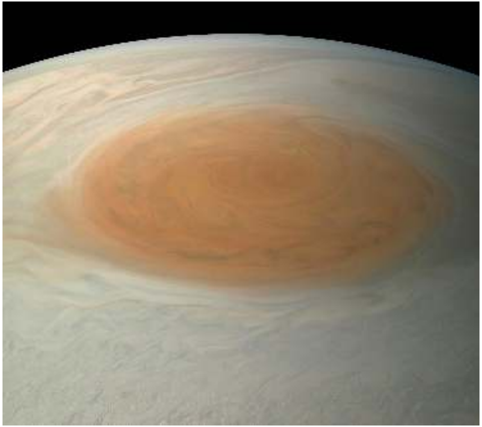
Venus: Super-Rotation

- Venus's surface rotates once every 243 Earth days (retrograde)
- Cloud-top winds: ~100 m s^{-1}, circling the planet in ~4 days
- Atmosphere rotates **60× faster than the planet**
- This is **atmospheric super-rotation**

A major puzzle

Super-rotation requires angular momentum transport **upward and equatorward**, against friction. Leading explanation: thermal tides + planetary-scale waves. The detailed mechanism remains an active area of research.

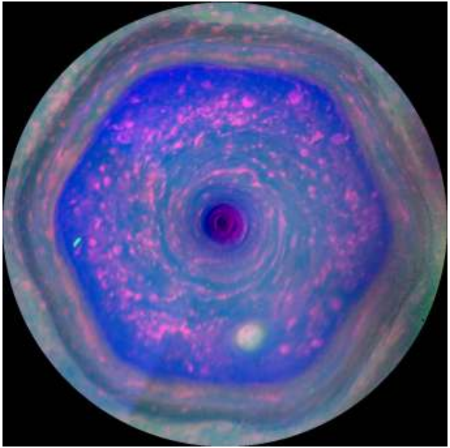
Jupiter: The Great Red Spot



Credit: NASA/JPL-Caltech/SwRI/MSSS (Juno)

- **Largest storm in the solar system**
- Anticyclonic vortex, larger than Earth
- Winds: ~120 $\mathrm{m\,s^{-1}}$ at periphery
- First recorded 1664; continuous observations since ~1830
- Confined between two opposing zonal jets
- Sustained by absorbing smaller vortices + deep $\mathrm{H_2O}$ latent heat
- Slowly **shrinking** over the past century

Saturn: The Hexagonal Jet Stream



Credit: NASA/JPL-Caltech/SSI (Cassini, 2013)

- Persistent hexagonal pattern at ~78°N
- Diameter: ~30,000 km (> Earth's diameter)
- Discovered by Voyager (1981), imaged by Cassini
- Explained as a stable **Rossby wave** with six-fold symmetry
- Resonance between jet speed, width, and Coriolis gradient

Saturn also has periodic **Great White Storms** (~every 30 yr; last: 2010).

Neptune: Extreme Weather on a Cold World

- Receives only ~1/900th of Earth's solar flux
- Yet has the **fastest winds** in the solar system: ~580 $m\,s^{-1}$
- Voyager 2 (1989): Great Dark Spot (since vanished), new storms formed
- Storms more transient than Jupiter's

Neptune radiates ~2.6× more energy than it receives from the Sun. Internal heat (gravitational contraction) drives convection and storms even with negligible solar heating.

A powerful reminder that solar flux alone does not determine weather.

7. Climate Evolution & the Faint Young Sun

Mars global dust storm (2018). Credit: NASA/JPL-Caltech/MSSS

Solar Luminosity Evolution

The Sun brightens as H → He increases the core mean molecular weight:

$$\frac{L(t)}{L_\odot} \approx \left[1 + \frac{2}{5}\left(1 - \frac{t}{t_\odot}\right)\right]^{-1}$$

- $t_\odot \approx 4.57$ Gyr (present age of the Sun)
- At formation ($t = 0$): $L/L_\odot \approx 0.71$
- The early Sun was ~**30% fainter** than today
- Even 4 Ga: still ~25% fainter

The Faint Young Sun Paradox

A 30% fainter Sun → $T_{\rm eff} \approx 255 \times (0.71)^{1/4} \approx 234$ K
With a modern greenhouse: surface well below freezing → **frozen ocean**.

But the geological record says otherwise:

- **Zircon crystals** (4.4 Ga): δ^{18}O ratios require liquid water
- **Pillow basalts and sedimentary rocks** (3.8 Ga): formed underwater
- **Stromatolites** (3.5 Ga): microbial mats requiring warm liquid water

Faint Sun should have frozen Earth, but geological evidence demands liquid water for >4 Gyr. This is the **faint young Sun paradox** (Sagan & Mullen, 1972).

Possible Solutions

- **Enhanced CO_2 greenhouse:** 10–1000× more CO_2 than today
  Plausible before the emergence of land plants
- **Methane greenhouse:** Biogenic CH_4 from early methanogens
  Long-lived in an anoxic (O_2-free) atmosphere
- N_2 **pressure broadening:** Thicker N_2 atmosphere (2–3× present)
  Enhances CO_2 and H_2O absorption bands
- **Lower albedo:** Less land area + fewer clouds → absorb more sunlight

Most likely: **combination** of elevated CO_2 and CH_4, regulated by the carbonate-silicate cycle.

The Mars Climate Puzzle

Mars at 1.52 AU → less than half of Earth's solar flux. With a faint young Sun: even worse.

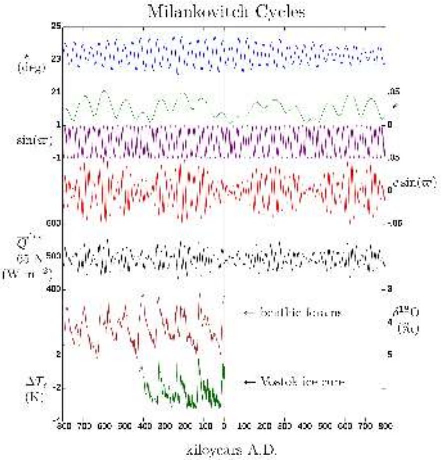
Yet the geological evidence for warm, wet conditions in the Noachian (>3.7 Ga) is strong:

- Extensive valley networks carved by flowing water
- Clay minerals from aqueous weathering
- Sedimentary deposits in ancient lake basins (e.g., Jezero crater)

Dense CO_2 alone is insufficient: it condenses into ice clouds that *increase* albedo.

Proposed solutions: Reducing gases (H_2, CH_4) from volcanism and water-rock reactions. Early Mars climate remains a **major open problem**.

Milankovitch Cycles: Orbital Forcing



Credit: Wikimedia, CC BY-SA

- **Obliquity:** 22.1°–24.5° on ~41 kyr
- **Eccentricity:** e = 0–0.06 on 100 and 400 kyr
- **Precession:** ~26 kyr
- Change seasonal/latitudinal insolation
- Pace glacial-interglacial cycles
- Match δ^{18}O in ocean sediments

Source: Hays et al. (1976); Laskar et al. (2004)

Climate Feedbacks

Climate stability depends on feedback mechanisms:

Ice-albedo feedback (positive):

Cooling → ice expands → higher albedo → less absorption → more cooling.
Can trigger **snowball Earth** (Neoproterozoic, ~700 Ma).

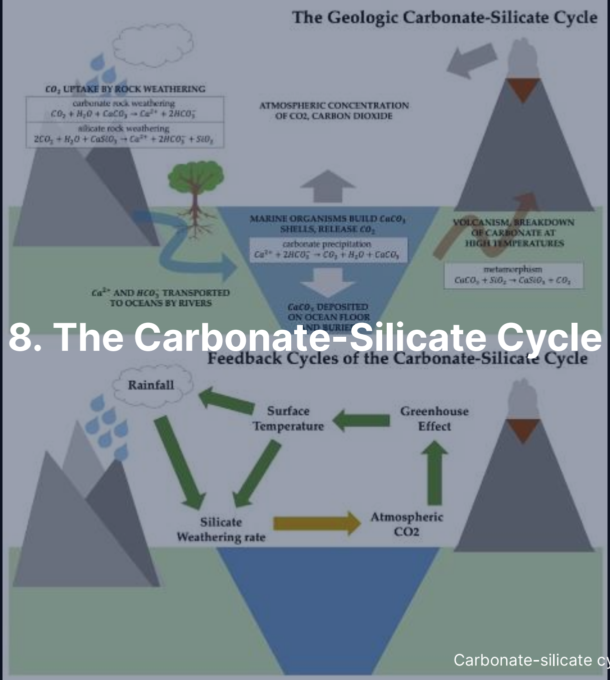
Water vapour feedback (positive):

Warming → more evaporation → $\mathrm{H_2O}$ is a greenhouse gas → more warming.
Approximately doubles CO_2 warming. If runaway: **Venus scenario**.

Cloud feedback (complex):

Low clouds: reflect sunlight (cooling). High cirrus: trap IR (warming).
Net effect: largest uncertainty in climate models.

8. The Carbonate-Silicate Cycle



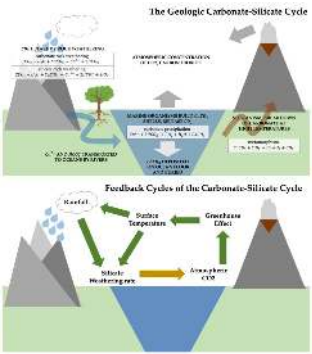
The Urey Reaction

Silicate weathering by atmospheric CO_2 dissolved in rainwater:

$$CaSiO_3 + CO_2 + H_2O \longrightarrow CaCO_3 + SiO_2 + H_2O$$

- CO_2 dissolves in rain → weak acid
- Acid reacts with silicate minerals at the surface
- Products: limestone ($CaCO_3$) and silica (SiO_2)
- Transported by rivers to the ocean → deposited as sediment
- Net effect: **draws** CO_2 **out of the atmosphere** (carbon sink)

Volcanic Outgassing: The Carbon Source



Credit: Wikimedia, CC BY-SA 4.0

- **Plate tectonics** closes the cycle
- Carbonate ocean floor is subducted into the mantle
- High *T* and *P* decompose carbonates → release $\mathrm{CO_2}$
- $\mathrm{CO_2}$ returned to the atmosphere via **volcanic outgassing**
- This is the long-term carbon **source**

The cycle balances sink (weathering) against source (volcanism).

The Negative Feedback Thermostat

Weathering rate depends strongly on temperature (Arrhenius-type):

1. **Planet warms:**
   More rainfall + faster reactions → weathering increases
   → more CO_2 removed → greenhouse weakens → **cools down**
2. **Planet cools:**
   Less rainfall + slower reactions → weathering decreases
   → volcanic CO_2 accumulates → greenhouse strengthens → **warms up**

This thermostat operates on ~0.5 Myr timescales. It is the primary reason Earth has maintained habitable temperatures for >4 Gyr despite a 30% increase in solar luminosity.

Venus & Mars: Failed Thermostats

The cycle requires **liquid water** (to weather rocks) and **active volcanism** (to recycle carbon):

Venus:

- Lost surface water (runaway greenhouse + H escape)
- No water → no silicate weathering
- Volcanic CO_2 accumulated unchecked
- Result: 92 bar CO_2 atmosphere

Mars:

- Interior cooled, volcanism ceased (~3 Ga)
- No volcanic CO_2 source
- Remaining CO_2 drawn down by weathering + lost to space
- Result: thin 6 mbar atmosphere

Habitability requires not just the right temperature, but the right **geological activity** to sustain it over billions of years.

9. Recent Advances & Outlook

Saturn’s hexagonal jet stream. Credit: NASA/JPL-Caltech/SSI

The Runaway Greenhouse

A planet that absorbs too much stellar flux can lose its oceans entirely.

- Water vapour is a strong greenhouse gas
- Warmer surface → more evaporation → more greenhouse → hotter surface
- Outgoing longwave radiation asymptotes to ~280–350 W/m^2 (Simpson-Nakajima limit)
- If absorbed stellar flux exceeds this, the positive feedback is unstoppable
- Oceans fully evaporate; H_2O photolysed by UV; H escapes hydrodynamically
- End state: dry, CO_2-rich atmosphere (Venus today)

Source: Nakajima et al. (1992); Goldblatt et al. (2013)

Ice-Albedo Feedback: The Ultimate Amplifier

A positive feedback loop that can destabilise climate:

T drops → ice cover grows → albedo rises → less solar absorption → T drops further

- Ice and snow reflect ~60–90% of incident sunlight vs. ~10–30% for ocean and land
- For Earth: the feedback is strong enough to create a **tipping point**
- If global ice cover reaches ~30% (down to ~30° latitude), the feedback becomes irreversible
- Runaway cooling drives the planet into a **snowball state**
- Amplification factor: $\Delta T_{\text{total}} \approx 2\text{–}3 \times \Delta T_{\text{forcing}}$ on modern Earth
- The dominant concern for modern climate: loss of Arctic sea ice accelerates warming regionally

Source: Pierrehumbert (2010); North et al. (1981)

Snowball Earth

Credit: Wikimedia Commons, public domain

- Earth may have gone through **two snowball states** in the Neoproterozoic (~720 and 635 Ma)
- Triggered by ice-albedo runaway when the ice line reaches ~30° latitude
- Surface temperatures drop below -50°C
- Volcanic CO_2 accumulates over ~10–100 Myr
- Greenhouse overwhelms albedo; ice melts rapidly
- Correlated with the rise of complex life shortly afterward

Venus' Super-Rotation: Thermal Tides

How does Venus' atmosphere rotate 60 times faster than the surface?

- Surface rotation: 243 days, retrograde
- Cloud-top winds: 4 days, ~100 m/s
- Drives the atmosphere equatorward and upward from the surface
- **Thermal tides:** solar heating of the cloud layer creates day-night pressure gradients
- These generate atmospheric waves that transport angular momentum upward
- **Planetary-scale waves** redistribute angular momentum meridionally
- Combined effect: upward-equatorward flux sustains super-rotation
- Detailed mechanism still debated; Akatsuki measurements ongoing

Source: Sánchez-Lavega et al. (2017)

Quantifying the Weathering Feedback

A simple model for the carbonate-silicate thermostat:

$$W(T) = W_0 \exp\left(\frac{T - T_0}{T_e}\right)$$

where W is the weathering rate, $T_e \approx 10$ K is the e-folding temperature.

- Doubling weathering requires ~7 K warming
- Halving it requires ~7 K cooling
- In steady state, $W = V$ (volcanism) sets atmospheric pCO_2
- Perturbation analysis: climate damping timescale ~0.5 Myr
- Works only if liquid water and active volcanism coexist

Source: Walker et al. (1981); Pierrehumbert (2010)

Neptune's Extreme Weather

Despite receiving only 1/900$^{\mathrm{th}}$ of Earth's solar flux, Neptune has the fastest winds in the solar system.

- Wind speeds up to 580 m/s (~2100 km/h)
- **Great Dark Spot** discovered by Voyager 2 (1989)
- Disappeared by 1994 (HST observations); new dark spots have since formed
- Energy source: internal heat, not sunlight
- Neptune radiates ~2.6× more power than it absorbs from the Sun
- Heat from slow contraction + possible compositional settling in the interior
- Transient storms contrast with Jupiter's long-lived GRS

Source: Hammel et al. (1989); Pearl & Conrath (1991)

The Venus Phosphine Debate

In 2020, a team reported tentative detection of phosphine (PH_3) in Venus' clouds.

- Greaves et al. (2020): ~20 ppb PH_3 at 53–60 km altitude
- No known abiotic source in Venus' atmosphere could explain this abundance
- Sparked biological speculation: could PH_3 be produced by hypothetical Venusian microbes?
- Subsequent reanalyses: the signal is much weaker (Villanueva et al. 2021)
- Current status: unconfirmed, controversial
- Motivated new Venus missions: DAVINCI (NASA), EnVision (ESA), VERITAS (NASA)

Source: Greaves et al. (2020), *Nat. Astron.*; Villanueva et al. (2021)

Dragonfly: A Rotorcraft for Titan

NASA's Dragonfly (launch 2028, arrival 2034) will explore Titan's surface.

- Nuclear-powered rotorcraft, the first of its kind for planetary exploration
- Mass: ~450 kg; can fly ~10 km between science stops
- Will land in the Selk impact crater region
- Sampling organic-rich dunes for mass spectrometry
- Investigating prebiotic chemistry: does the combination of complex organics and transient impact-melt water produce biomolecules?
- 2.7 year primary mission, visiting 20+ sites

Source: Turtle et al. (2019)

JWST: Weather on Distant Worlds

JWST is transforming our view of exoplanet atmospheres.

- WASP-39 b: first clear detection of CO_2 in an exoplanet atmosphere (2022)
- WASP-18 b: sub-micron wavelength thermal emission shows dayside-to-nightside heat redistribution
- WASP-107 b: first firm detection of SO_2 $\Rightarrow$ photochemistry in action
- Measurements of terminator cloudiness (east-west asymmetry)
- Searching for weather patterns, circulation signatures
- First steps toward characterising habitable-zone rocky planet atmospheres

Source: The JWST Transiting Exoplanet Community ERS Team (2023)

Recent Advances: Early Mars Climate

- Updated 3D climate models with reducing gases (H_2, CH_4) can warm early Mars
- Wordsworth et al. (2021): collisionally-induced absorption by H_2-CO_2 is a strong greenhouse effect at high CO_2 pressures
- Impact-triggered warming events: large impactors deliver heat and deflect vapour into the atmosphere
- Localised glaciation models: the “icy highlands” picture
- Episodic vs. sustained warming: still debated
- Perseverance (Jezero crater, 2021–) investigating the aqueous record

Source: Wordsworth (2016); Wordsworth et al. (2021)

Summary

1. Clouds form when rising air cools past the **saturation vapour pressure** at the LCL
2. The Clausius-Clapeyron equation gives $P_{\mathrm{sat}}(T) \propto \exp(-L_v/R_vT)$; governs all clouds
3. Cloud species vary: $\mathrm{H_2O}$ (Earth), $\mathrm{H_2SO_4}$ (Venus), $\mathrm{NH_3/H_2O}$ layers (Jupiter)
4. Atmospheric circulation: Hadley cells, Coriolis deflection, jet streams
5. Rotation rate sets number of circulation cells (1 on Venus, 3 on Earth, ~15 on Jupiter)
6. Weather: Mars dust storms, Venus super-rotation, Jupiter GRS, Saturn hexagon
7. Faint young Sun paradox: early Sun 30% fainter, yet Earth had liquid water
8. Carbonate-silicate cycle: Earth's climate thermostat; failed on Venus and Mars

Questions?

Next: **Lecture 7. Planetary Surfaces: Geology, Geomorphology, & Geophysics**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`